

AnyBlox: Decoupling Storage Formats from Data Processing Systems

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Abstract

Open storage formats in data lakes have separated data from the systems that process it, reducing vendor lock-in. However, reading a format still requires a decoder for that format to be present in every consuming system. For N systems and M formats, this results in $N \times M$ independent decoder implementations, which raise maintenance costs, increase the risk of security vulnerabilities, and discourage the adoption of new encodings. To address this, Gieniec et al. (2025) propose AnyBlox, a framework that moves the decoder into the data itself: each dataset ships with its own decoder as sandboxed WebAssembly bytecode. With this approach, any engine that implements the framework once can read a format without prior knowledge, reducing the integration burden to $N + M$. The sandbox overhead remains negligible in end-to-end workloads. Furthermore, more efficient encodings outperform established formats in both scan latency and compression, directly addressing the maintenance burden and format ossification caused by the $N \times M$ problem.

1 Introduction

For decades, database systems functioned as tightly coupled units, with a single system controlling storage format, processing logic, and data access. External systems submitted queries and received results; internal operations remained hidden. With complete control over the physical layout, systems can optimise data storage and processing within their own boundaries. This model was sufficient as long as data did not cross system boundaries [19].

However, the volume and variety of collected data have grown drastically. Data now arrives continuously from a growing number of sources, in formats suited to their respective domains rather than to any particular database system. As a result, the database landscape has shifted from a single system controlling all aspects of data management to a heterogeneous, modular ecosystem of specialised systems, each handling a distinct part of the data pipeline using a shared storage layer. Open file formats such as Apache Parquet¹ and Apache ORC², stored in data lakes [3], are widely adopted in this context. Beyond these, data scientists increasingly analyse data using domain-specific encodings from fields such as high-energy physics and bioinformatics [8].

Independently, research in data compression and encoding continues to produce formats that surpass established standards for storage efficiency and query throughput. Yet, these advances rarely appear in production systems, because each system must independently implement support for every new format it wants to read [8].

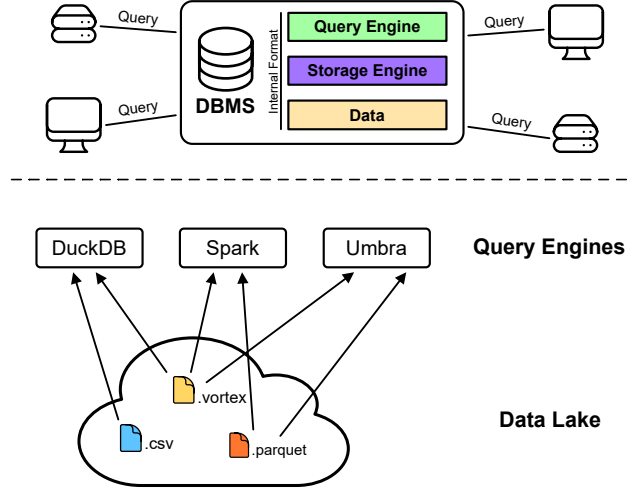


Figure 1: (Top) A monolithic DBMS controls its query engine, storage engine, and internal format; clients interact only through queries. (Bottom) In a data lake, multiple query engines access a shared pool of open file formats. Since each must implement a decoder for each format, N engines and M formats require $N \times M$ independent implementations.

Both trends lead to the same structural problem. Codd’s principle of physical data independence [6] assumes a system controls its storage format and shields applications from physical details. In a shared storage layer, this does not apply. The data producer determines the format, so any system that reads a format must implement its own decoder. With N systems and M formats, there are $N \times M$ independent implementations to develop and maintain.

This results in higher implementation costs, greater maintenance effort, and increased security risks, as each implementation may contain bugs. For example, a high-severity vulnerability in PostgreSQL’s³ JSON support went undetected for over five years [17]. This initial hurdle discourages systems from supporting new formats altogether.

The consequence is a self-sustaining cycle: systems do not adopt a new file format because its user base is too small, and the user base remains small because the format lacks support. Gieniec et al. (2025) identify this as a core structural problem because an encoding must be popular enough to justify support, but cannot gain popularity without broad support.

¹Apache Parquet, <https://parquet.apache.org> (accessed: May 22, 2026)

²Apache ORC, <https://orc.apache.org> (accessed: May 22, 2026)

³PostgreSQL, <https://www.postgresql.org> (accessed: May 22, 2026)

This dynamic leads to format ossification, where datasets use outdated formats just to stay compatible, even when better encodings exist [11]. Changing formats is difficult because it risks breaking compatibility with many systems that use older specifications.

But what could solve this combinatorial integration burden? An abstraction layer between storage formats and processing systems reduces the $N \times M$ problem to $N + M$. Each system implements the interface once, and each format provides a single implementation against it. Gienieczko et al. (2025) propose exactly this and present AnyBlox as their realisation.

2 Self-Decoding Data: The AnyBlox Design

A future-proof data access solution must simultaneously satisfy two conceptual requirements. First, it must introduce an abstraction layer that decouples format internals from processing systems and system internals from decoders, reducing the $N \times M$ integration burden to $N + M$. Second, this abstraction cannot preclude direct file access, since routing data through an intermediary process introduces transfer overhead that can dominate total runtime. These two goals are in tension: an abstraction layer must reside somewhere, and each possible location either breaks direct access or reintroduces format-specific coupling [8].

Existing approaches each resolve this tension in favour of one side. Native integration preserves direct file access and achieves the best performance through tight coupling with host internals, but requires a separate decoder implementation for each format and system, thereby reproducing the $N \times M$ problem. Extension mechanisms reduce the per-format implementation effort, but each host exposes its own extension API, so a decoder written for system A cannot be reused in system B without a near-complete rewrite. Isolated extensions, such as containerised user-defined functions [18] or remote cloud user-defined code execution solutions, provide both isolation and portability, but introduce a data transfer bottleneck between the host and the isolated decoder process [8].

None of these approaches satisfies both requirements at once. Native integration, extensions, and isolated extensions each forgo one of them, while static verification targets both but is held back by the limitations of current verifier technology [8]. Gienieczko et al. (2025) therefore argue that only one location for the decoder remains: inside the data file itself. Packaging the decoder with the data eliminates the transfer bottleneck, since decoder and data reside in the same address space, while the abstraction is preserved because the host need not implement any format-specific logic. This is the core idea behind self-decoding data and the design premise of AnyBlox.

2.1 WebAssembly as the Decoder Runtime

A self-decoding dataset requires a decoder runtime that satisfies four criteria: **Portability**, **Security**, **Performance** and **Extensibility**.

Building on these criteria, Gienieczko et al. (2025) identify WebAssembly (Wasm) as a runtime that satisfies all four simultaneously. Wasm is a portable low-level bytecode and virtual machine specification, designed as an abstraction over modern hardware rather than a commitment to any specific programming model, making it language-, hardware-, and platform-independent [9].

2.1.1 Portability. A decoder implemented in Wasm can operate across a broad set of host architectures, including browsers, mobile devices, embedded systems, and server workstations. With AnyBlox, a format author can write and compile a decoder once and execute it on any host through libanyblox, regardless of the underlying hardware or operating system [8].

2.1.2 Security. The WebAssembly specification provides machine-verifiable guarantees of memory safety and isolation [21]. Building on this foundation, AnyBlox reinforces these guarantees by being implemented predominantly in safe Rust, ensuring memory and thread safety. Furthermore, all unsafe code is confined to a single module that handles Wasm memory maps, making it straightforward to inspect [8].

2.1.3 Performance. Wasm is JIT-compiled to the host's native instruction set at runtime, enabling near-native performance. Additionally, AnyBlox avoids unnecessary data copying through a custom memory management scheme, which improves efficiency. As observed by Gienieczko et al. (2025), AnyBlox further benefits by passively inheriting advances in Wasm compiler technology, since all JIT and sandboxing details are encapsulated within libanyblox and not exposed in the public API [8].

2.1.4 Extensibility. Finally, in terms of extensibility, since most general-purpose programming languages provide a Wasm toolchain, format authors can implement decoders in their language of choice. The main practical obstacle is SIMD instructions: converting platform-specific SIMD intrinsics to Wasm's instruction set requires manual developer effort. However, since Wasm defines SIMD at the bytecode level, the resulting modules remain fully portable across architectures [8].

2.2 Reading a Dataset: From File to Query Pipeline

To read an AnyBlox dataset, the Host opens the .any file and reads its metadata during query planning. The metadata specifies the schema, row count, an estimate of the decoded data size, and recommended batch size, which are used to partition workloads across threads and to schedule scans before decoding.

The Host passes metadata and the .any file to libanyblox, which extracts, JIT-compiles, and caches the Wasm Decoder module. Later uses of the same Decoder skip compilation. AnyBlox then sets up the Decoder's execution environment and returns a job handle to the Host.

This procedure describes two distinct AnyBlox modes. In standalone mode, the dataset resides in its own .any file with a thin metadata layer. AnyBlox reliably encodes the schema and row count; further hints, such as an estimated decoded size or a minimum recommended batch size, may be present but are treated as optional. In embedded mode, AnyBlox is incorporated within an existing host format (such as Parquet). Here, the Host uses its own decoding library, and the embedded path inherits the Host format's statistics, column-level filtering and row-group filtering. The decoding pipeline described below is identical in both modes. The key difference is the metadata available to the Host, which is relevant for predicate pushdown (Section 3.3).

From this point, the Host repeatedly requests arbitrary tuple ranges from that handle. Modern query engines process data in batches to reduce communication overhead between operators. Therefore, rather than decoding the entire dataset at once, the Host issues successive calls for ranges of tuples [14]. AnyBlox delegates each such request into the sandboxed Wasm module and returns the decoded result to the Host. The sole function through which this happens is `decode_batch` [8]:

- `i32 data`: pointer to the start of the encoded data in Wasm linear memory
- `i32 data_length`: length of the encoded data in bytes
- `i32 start_tuple`: ID of the first tuple to decode
- `i32 tuple_count`: number of tuples to decode
- `i32 state`: pointer to the state page in Wasm linear memory
- `i64 projection_mask`: bitmask specifying which columns to decode, allowing the Host to skip columns that are not needed for the current query

The `data` parameter references the encoded dataset in the Decoder’s execution environment. Copying the dataset into Decoder-accessible memory incurs an initialisation cost proportional to the dataset size. AnyBlox instead uses `mmap` to map the dataset file into the Decoder’s linear memory, avoiding copying. Physical memory is committed only upon first page access.

Wasm’s linear memory is a contiguous, byte-addressable memory region starting at address `0x0` that represents the entire address space accessible to the Decoder, isolated from the rest of the host process. AnyBlox controls its layout. The first pages hold the Decoder module’s code, stack, and static data (typically a few megabytes). The encoded dataset is mapped via `mmap` into the first page after this initial region, with the state page directly following the dataset; Decoder allocations expand from the following pages. The `data` parameter thus points to the start of the mapped dataset. Reusing this linear memory region across jobs on the same thread eliminates the need for repeated initialisation.

The `state` parameter points to a page within the linear memory that is zero-initialised at the start of every job. A Decoder can use it to cache metadata or memory allocations across successive `decode_batch` calls. For example, a run-end encoding Decoder that performs a binary search on the first call can store its current position on the state page and resume directly on the next call, skipping the search entirely. Because the state page resets at the start of each job, Decoders are idempotent: identical parameters applied to the same dataset always produce the same results.

Once decoding is complete, `decode_batch` returns a pointer to an Apache Arrow Record Batch within the linear memory. Arrow⁴ is a columnar in-memory format developed by the Apache Software Foundation. Rather than storing data row by row, it organises each column as a contiguous array in memory, allowing query engines to efficiently process large amounts of data without unnecessary memory accesses. A Record Batch is a collection of such arrays, all of the same length, representing a horizontal slice of a table. Arrow standardises the representation of common SQL types (e.g., integers, floating-point numbers, strings, dates, and more) and is supported by a wide range of query engines. The choice of Arrow

as the uniform output format of AnyBlox follows the same principle as other intermediate representations in systems design, because by agreeing on a common language between two components, the number of required translations reduces from $N \times M$ to $N + M$. Every Decoder, regardless of the underlying encoding scheme, produces an Arrow Record Batch, and every Host that speaks Arrow can consume it without any AnyBlox-specific logic. AnyBlox translates the Wasm-internal pointer into a native address and passes the batch to the Host, where it enters the query pipeline. The degree of integration varies by system: DataFusion [12] uses Arrow as its primary internal representation throughout, enabling zero-copy end-to-end transfers. DuckDB [16] and Spark⁵ do not use Arrow internally, but both have built-in support for reading Arrow batches and converting them into their respective internal representations on the fly. This conversion logic is part of the mainline code of both systems and requires no AnyBlox-specific adaptation [8].

3 Discussion

The preceding sections introduced the AnyBlox architecture and its decoding model. Whether this design is practical depends on three questions:

- (1) What overhead does the Wasm sandbox introduce?
- (2) How much effort does integrating the framework into an engine require?
- (3) Where does the approach reach its limits?

Gienieccko et al. (2025) report benchmarks and integration experience that cover both favorable and unfavorable cases for AnyBlox.

3.1 Performance Evaluation

Sandboxing a decoder inside Wasm adds a layer of indirection between the data and the query engine. Gienieccko et al. (2025) measure the resulting overhead directly. They run TPC-H at scale factor 20 in DataFusion [12] and vary the representation of `lineitem`, the largest table in the benchmark and therefore the one whose scan dominates the runtime, across three configurations:

Parquet read by DataFusion’s native reader with Snappy⁶ compression;

AnyBlox Vortex a Vortex file decoded inside the AnyBlox sandbox;

Native Vortex the same Vortex decoder compiled natively without any sandbox.

The first variant isolates the effect of the format; the latter two isolate the effect of the sandbox alone. Predicate pushdown was disabled throughout, so the numbers reflect scan and operator cost only. Vortex⁷ is a recent columnar format.

The benchmark yields two results, summarized in Figure 2. Both Vortex variants achieve over $2\times$ lower scan latency than Parquet while also reaching a better compression rate. For the Parquet reader, decoding accounts for nearly half of the processing time, while Vortex spends around 80% of its processing time in relational operators. The overhead of the sandbox itself is small. Sandboxed Vortex decoding adds 5% to the scan latency over native decoding, which amounts to a 1% increase in overall query latency and which

⁴Apache Arrow, <https://arrow.apache.org> (accessed: June 2, 2026)

⁵Apache Spark, <https://spark.apache.org> (accessed: June 2, 2026)

⁶Snappy, <https://github.com/google/snappy/> (accessed: June 15, 2026)

⁷Vortex, <https://vortex.dev/> (accessed: June 15, 2026)

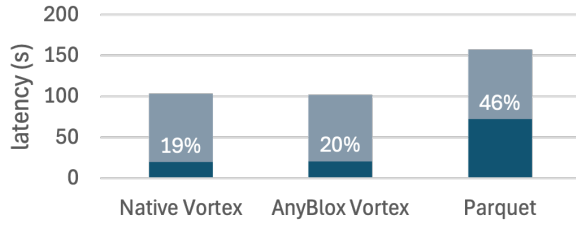


Figure 2: Total query latency (median, summed over 16 TPC-H queries at scale factor 20) in DataFusion, split into scan time (dark) and operator time (light). Percentages denote the scan share of total latency. Data source: [8].

the authors report as within measurement noise. Across this end-to-end workload, the cost of the sandbox is therefore negligible, and AnyBlox provides DataFusion with an encoding that outperforms its own native Parquet reader [8].

The microbenchmarks show where the overhead of the sandbox becomes visible. FSST [5] is a lightweight string compression scheme. Gienieczko et al. (2025) benchmark the decoding of three FSST-compressed string columns from the Taxpayer dataset of the Public BI Benchmark [7]. The 1.7 GB AnyBlox file expands into 2.5 GB of Arrow batches in memory. In this isolated scenario, the Wasm Decoder is over 43% slower than the same decoder compiled natively. The cause is architectural rather than incidental: the FSST decoder relies heavily on 8-byte reads, but Wasm uses a 32-bit address space by default, so each 8-byte read is emulated with two 4-byte reads. The authors describe this as a fundamental limitation of the current design. The 64-bit Wasm proposal would remove it, but it is incompatible with the software fault isolation mechanism on which AnyBlox’s security guarantees rest, since that mechanism assumes a Decoder can address at most 8 GiB of memory [8].

Taken together, the benchmarks place the cost of the sandbox in perspective, though the evidence differs in strength across the two cases. For a complex decoder such as Vortex, the overhead is measured directly: in the end-to-end DataFusion workload it stays within measurement noise, and the more efficient encoding lets AnyBlox outperform the native Parquet reader. For the FSST worst case, the slowdown appears clearly in the microbenchmark, but its effect on full queries is not measured directly. Since the scan accounts for only about a fifth of processing time in the DataFusion workload, a decoding slowdown can affect only that share of the runtime. The microbenchmark therefore isolates the decoder under the conditions least favorable to it.

3.2 Integration Effort

The second claim AnyBlox makes is portability, meaning an engine can support the framework without a large implementation effort. Gienieczko et al. (2025) integrated AnyBlox into four systems spanning different execution paradigms (Umbra [15], DuckDB, DataFusion and Spark), all done by a single programmer with no prior experience in any of the host codebases. Table 1 lists the resulting effort. No integration required more than 1305 lines of code, and each one only added a new scan operator while leaving the

Table 1: Effort of integrating AnyBlox into each host system. Data source: [8].

System	Lines of code	Person-days
DataFusion	461	2
DuckDB	586	3
Spark	756	15
Umbra	1305	10

rest of the engine, namely its optimizer, operators and execution layer, unchanged [8].

The authors caution that lines of code and integration time are not rigorous measures of software complexity, so these figures indicate rather than prove that the framework ports easily. With that caveat, they are the empirical counterpart to the $N \times M$ argument: a one-time integration lets an engine read any format providing a decoder.

3.3 Limitations

Two of AnyBlox’s limitations follow from the design itself rather than from the current implementation. The first concerns predicate pushdown. Predicate pushdown evaluates filter conditions during the scan phase, so that rows failing a condition are discarded before they are decoded and passed to operators such as aggregation or joins. Decoding is computationally expensive, so prior filtering avoids processing data that would be discarded anyway [19].

Standalone AnyBlox provides no predicate pushdown API for row filtering. It supports projection pushdown, which restricts a scan to the requested columns, but not the filtering of individual rows. The reason is a deliberate design choice: filtering individual rows in a format-agnostic framework requires an equally format-agnostic and system-independent predicate expression format, which AnyBlox does not define. This reflects the same design philosophy visible elsewhere in the framework, that statistics and indexes should be decoupled from the storage format and managed by the query engine or a dedicated acceleration layer. The authors point to Substrait⁸ as ongoing industry work toward a standardized predicate expression format, which could close this gap in a future revision [8].

This limitation is confined to standalone use. When an AnyBlox encoding is embedded into a host format such as Parquet at the row-group or column level, the host’s existing row-group and column filtering continues to apply, so the embedded path retains the pushdown that the host already performs [8].

The second limitation concerns memory. Standalone AnyBlox maps the entire dataset into the Wasm linear memory region via `mmap` before decoding begins. The current implementation uses `wasm32`, which provides a 32-bit address space. Each mapped file is therefore limited to 4 GB. The authors note that file sizes in typical analytical deployments are commonly kept well below 4 GiB to facilitate cheap updates, so the restriction may have limited impact in practice. A migration to `wasm64` would remove the limit, since Memory64 has been part of the WebAssembly standard since

⁸Substrait, <https://substrait.io/> (accessed: June 15, 2026)

2025⁹. However, the performance impact of this migration in the context of AnyBlox is not yet studied, and wasm64 is incompatible with the software fault isolation mechanism on which AnyBlox’s security guarantees rest. The mmap requirement also means that column projections cannot be pushed to remote files, because the entire dataset must reside in memory before the decoder is invoked, which is a relevant constraint in data lake deployments where data typically resides on object stores. Partial remote reads via page-fault handling are sketched as future work but not yet implemented [8].

4 Conclusion

Reading an unfamiliar storage format normally requires a reader written for that format inside every system that wants to consume it. AnyBlox removes this requirement by moving the reader into the data. A decoder travels with the dataset as sandboxed WebAssembly bytecode, and any engine that implements a single scan interface can decode the format without knowing it in advance. The cost of this design is the overhead of the sandbox, which the measurements of Gienieccko et al. (2025) show to be negligible for the end-to-end workloads tested.

The authors frame the underlying problem as an $N \times M$ problem: N systems each supporting M formats incur $N \times M$ implementation and maintenance effort [8]. An abstraction layer between the two sides reduces this to $N + M$, since each engine implements the interface once and each format provides one decoder. The authors draw this analogy explicitly from compiler construction, citing the language-independent intermediate representation of LLVM [13], which decouples M language frontends from N target backends through a shared representation. AnyBlox applies the same decoupling between storage formats and query engines. The comparison is an analogy rather than evidence: it motivates the design but does not establish that the result transfers from compilers to data formats.

This decoupling is not an isolated idea but part of a broader move toward composable database systems, away from large, monolithic designs. The authors place AnyBlox alongside several efforts that each modularise a different layer, among them a query engine built on composable design principles [12], execution-plan frameworks made extensible through sub-operators [4, 10], portable query optimizers [1, 2], and system-independent persistence mechanisms [20]. AnyBlox contributes the storage-format layer to this picture, and one of these efforts connects directly to it: the predicate pushdown limitation discussed in Section 3.3, where standalone AnyBlox deliberately defines no row-filtering predicate format, is a gap that Substrait, an emerging cross-engine standard for query plans, could close. The components are complementary, each standardising one interface between otherwise independent systems [8].

As a precedent that such modular frameworks can succeed, the authors invoke LLVM, which reshaped the design of compilers by establishing a shared, open representation [8, 13]. AnyBlox shows that the same principle could work for storage formats. Its evaluation demonstrates that a format-agnostic decoding layer can be built with low overhead and integrated into engines across different execution paradigms with modest effort. The LLVM comparison

remains an analogy, and the evidence so far comes from a single evaluation by the original authors, so whether modular stacks come to displace integrated systems is a question that broader adoption and independent study will decide. What the paper establishes is that the decoding layer is no longer the obstacle.

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⁹WebAssembly Memory64, <https://github.com/WebAssembly/memory64> (accessed: June 15, 2026)

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